

July 25, 2006



U.S. Department
of Transportation

400 Seventh Street, S.W.
Washington, D.C. 20590

**Pipeline and Hazardous
Materials Safety Administration**

DOT-SP 10277
(TENTH REVISION)

EXPIRATION DATE: June 30, 2010

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Amtrol, Inc.
West Warwick, RI
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, marking, sale and use of a non-DOT specification cylinder conforming in part with the DOT Specification 4BA for the transportation in commerce of certain Class 2 materials. This special permit provides no relief from the Hazardous Materials Regulations other than as specifically stated herein.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 180.209(e), § 173.304 and § 175.3 in that a non-DOT specification packaging is not authorized except as specified herein.
5. BASIS: This special permit is based on the application of Amtrol Inc., dated July 18, 2006 submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR § 172.101): See Appendix A of this special permit.

7. PACKAGING(S) and SAFETY CONTROL MEASURES:

a. PACKAGING: Packaging prescribed is a non-DOT specification welded steel cylinder conforming with DOT Specification 4BA (§ 178.51) in effect on October 1, 1995, except as follows:

§ 178.51-1 Compliance.

Each cylinder must conform with this special permit in all details and with § 178.51 unless otherwise noted herein.

§ 178.51-2 Type, size and service pressure.

(a) Type. Cylinders must be of cylindrical shape, with integrally formed spherical or elliptical heads. Cylinder ends closed by spinning process are not authorized. Cylinders must be fabricated from ASTM A 620 class 1 deep drawing quality steel with seamless cylindrical shells with integrally formed heads, and joined by welding of one circumferential seam. Brazed joints for attachments when appropriate are authorized.

(b) Water capacity not to exceed 50 pounds.

(c) Service pressure: Not to exceed 400 pounds per square inch gauge.

§ 178.51-4 Duties of inspector:

(a) * * *

(b) The inspector must verify compliance with the material specification requirement in this special permit by obtaining a certified chemical analysis of each heat of the specified steel (ladle analysis is acceptable). The certificate from the manufacturer, thereof, giving sufficient data to show compliance with the requirement is acceptable when verified by check analysis on one container from each lot.

(c) * * *

(d) * * *

§ 178.51-5 Steel.

(a) The composition of the steel authorized is limited to ASTM SA 620 class 1 designation aluminum killed, deep drawing quality steel with carbon content not to exceed 0.10%. Certification of the chemical composition and mechanical properties for each heat of steels used for production is required. The steel analysis must conform with the following:

Element	Weight %
Carbon	0.10 (max)
Manganese	0.50 (max)
Phosphorous	0.025 (max)
Sulphur	0.035 (max)
Aluminum	0.010 (min)

§ 178.51-8 Manufacture.

(a) Cylinder shells must be of seamless construction manufactured by the deep drawing method with integrally formed heads; dirt and scale to be removed as necessary to afford proper inspection; fissures or other defects that are likely to weaken the finished cylinder appreciably are not acceptable. Irregularities such as draw marks, scratches, pits, etc. should be held to a minimum. If the cylinder is not originally free of such defects or does not meet the finish requirements, the surface may be machined or otherwise treated to eliminate these defects. Metal removal for any purpose other than removal of isolated defects and threading must be done prior to the hydrostatic test. The thickness of the treated areas must be measured and may not be less than the minimum prescribed thickness. Cylinder end contour must be hemispherical or ellipsoidal (axis ratio of 2:1) with concave side to pressure. Shape, thickness, and strength of the cylinder bottom and side wall adjacent to the bottom must be such that failure during the cycle pressure test occurs in the sidewall of the cylinder.

(c) Delete

(d) Welded joints must have strength equal to or greater than the minimum strength of the shell material in the finished cylinder.

§ 178.51-10 Wall thickness.

(a) The minimum wall thickness must be not less than 0.078 inches. The thickness of the top and bottom heads must be no less than the cylinder sidewall thickness. In any case the maximum stress in the cylinder wall at test pressure, calculated by the formula in § 178.51-10 (b), may not be greater than 35,000 psi.

§ 178.51-11 Heat treatment.

Does not apply

§ 178.51-14 Hydrostatic test:

(a) * * *

(b) Each completed cylinder must be tested at an internal pressure of at least two times the rated service pressure, and must be held at that pressure for a minimum of 30 seconds.

(1) Cylinders that show excessive deformation and defects must be rejected.

(2) Volumetric expansion measurement to compliment the test is required on one cylinder in each lot of 200. The cylinders in each lot must be of identical size, design, construction, material specification, finish, and quality.

(3) Each cylinder in a lot must be tested for leaks by immersing the pressurized cylinder in a water bath. Other methods which are equivalent or better are permitted. If any cylinder leaks, or if there is evidence of distortion or defects while under test, that cylinder must be rejected.

(c) One cylinder selected at random from each lot of 200 or less successively produced, must be hydrostatically tested to burst. The observed pressure at burst must be at least 4 times the service pressure.

§ 178.51-15 Physical test.

(a) Determine yield strength, tensile strength, elongation, and Charpy impact energy at room temperature (70°F). Required on two specimens for tensile test and three specimens for Charpy impact test, taken from a qualification cylinder sidewall.

The cylinder must be from the first lot produced from each heat steel.

(b) * * *

(c) * *

(d) Flattening test.

One cylinder taken from the beginning of each production lot of 200 cylinders must be subjected to flattening test as follows:

(i) The flattening test must be made on a cylinder that has been tested at test pressure.

(ii) A ring taken from the cylinder may be flattened as an alternative to testing a complete cylinder. The test ring may not include weld or heat affected zone.

(iii) The flattening test must be between 60 degrees included angle wedge shaped knife edges, rounded to a 0.5 inch radius.

(iv) When flattened fully, the outer surfaces of the test specimen cylinder must be parallel and apart not more than 6 times the wall thickness.

(v) If the test cylinder or ring, when fully flattened, shows evidence of surface cracks the cylinders of the lot represented by the test must be rejected.

§ 178.51-16 Elongation:

Delete

§ 178.51-18 Rejected cylinders.

(a) * * *

(b) If the cause for rejection of a lot is determinable and can be rectified, the lot may be qualified by the following procedure:

(i) Identify and record the cause for rejection of the lot. Perform the procedures eliminating the cause. Repair of the defective welds is permitted. Brazed joints may not be repaired.

(ii) Four additional cylinders, randomly selected from the lot, must be burst tested. If any of the cylinders fail at a pressure below two times the test pressure, the entire lot must be rejected.

§ 178.51-19 Marking.

(a) Marking of each cylinder must be in compliance with the requirements of § 178.51-19 except as follows:

(1) The cylinder must be marked "DOT-SP 10277-XXX", in lieu of "DOT-4BA" where XXX is the service pressure and may not exceed 400 psig.

(2) * * *

(3) * * *

(4) Date of manufacture,

(5) Date of first retest,

(6) Date of last permissible refill.

(b) * * *

(c) The above markings may be on the footing permanently attached to the cylinder.

(d) * * *

§ 178.51-20 Authorized Steel: As specified in section 178.51-5 in this special permit.

§ 178.51-21 Inspectors Report "The material used was Type ASTM A-620 class 1" must be substituted for "The material used was Type 1 authorized in Table I of Spec. No. 4BA".

b. The service life of these cylinders must not exceed 10 years from the date of manufacture.

c. Each cylinder must be marked with a warning label stating that it is illegal to refill the container after MM/YYYY; MM/YYYY being the month and the year of last permissible refill date corresponding to the requirement in the marking requirements in this special permit. The marking may be on a label, decal affixed to the surface of the packaging or by stenciling. The warning statement must be in English, legible, unobscured by other labels and attachments, and durable.

8. SPECIAL PROVISIONS:

a. Reports:

(i) Prior to the initial shipment of cylinders made in compliance with this special permit, a report of test results specified in § 178.51-14 and § 178.51-15 of this special permit must be submitted to the Office of Hazardous Materials Special Permits and Approvals (OHMSPA);

(ii) A copy of the Inspector's report for each of the first three lots produced must be submitted to the OHMSPA; and

(iii) All test reports including certificates of material specification compliance must be retained indefinitely or as long as the cylinders are authorized, and must be made available to a DOT representative upon request.

b. Filling limits must conform with the provisions of § 173.304. Under no circumstances are these cylinders to be filled to a pressure exceeding the marked service pressure at 70°F.

c. Before shipment, each charged cylinder must be tested for leaks by immersing fully in a water bath. Other methods for leak testing are authorized provided such testing is equivalent or better than the water bath test. Cylinders that show evidence of leakage may not be shipped.

d. At the end of 5 years from the manufacturing date, cylinders used in non-corrosive service must be subjected to visual examination of internal and external surfaces, as well as valve threads. Any cylinder showing evidence of corrosion or defects that can potentially effect cylinder integrity must be removed from service. Cylinders used in corrosive service must be retested in accordance with the requirements of § 180.209 as specified for DOT 4BA cylinders.

e. A cylinder must be condemned when it leaks, or when internal or external corrosion, denting, bulging, or evidence of rough usage exists to the extent that the cylinder is likely to be weakened appreciably, or when damaged by fire.

f. Offerors for transportation of the hazardous materials specified in this special permit may use the packaging described in this special permit for the transportation of such hazardous materials so long as no modifications or changes are made to the packages, all terms of this special permit are complied with, and a copy of the current special permit is maintained at each facility from which such offering occurs.

g. A current copy of this special permit in its current status and test procedures referenced therein must be maintained at each manufacturing facility at which the special permit cylinders are manufactured and must be made available to a DOT representative upon request.

h. Shippers using the packaging covered by this special permit must comply with all provision of this special permit, and all other applicable requirements contained in 49 CFR Parts 100-180.

i. New construction of cylinders after January 31, 2000 is not authorized.

j. Packagings permanently marked 'DOT-E 10277', prior to October 1, 2007 may continue to be used under this special permit for the remaining service life of the packaging or until the special permit is no longer valid. Packagings marked on or after October 1, 2007 must be marked 'DOT-SP 10277'.

k. Shipping papers displaying 'DOT-E 10277' may continue to be used until October 1, 2007, provided the special permit remains valid.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo aircraft only.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
 - o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

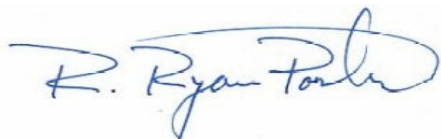
Each "Hazmat employee", as defined in § 171.8 who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU)- 'The Hazardous Materials Safety and Security Reauthorization Act of 2005' (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term 'exemption' to 'special permit' and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for Robert A. McGuire
Associate Administrator
for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, Department of Transportation, Washington, D.C. 20590. Attention: PHH-31.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at http://hazmat.dot.gov/sp_app/special_permits/spec_perm_index.htm Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: sln

Hazardous materials description - - proper shipping name	Hazard Class/ Division	Identi- fication Number	Packing Group
1,1,2-Tetrafluoroethane or Refrigerant gas R-134a	2.2	UN3159	n/a
1,2-Dichloro, 1,1,2,2- Tetrafluoroethane, or Refrigerant gas R-114	2.2	UN1958	n/a
Bromotrifluoromethane or Refrigerant gas R-13B1	2.2	UN1009	n/a
Chlorodifluorobromoethane or Refrigerant gas R12B1	2.2	UN1974	n/a
Chlorodifluoromethane and Chloropentafluoroethane mixture or Refrigerant gas R-502	2.2	UN1973	n/a
Chlorodifluoromethane or Refrigerant gas R-22	2.2	UN1018	n/a
Chloropentafluoroethane or Refrigerant gas R-115	2.2	UN1020	n/a
Chlorotrifluoromethane and Trifluoromethane mixture or Refrigerant gas R-503	2.2	UN2599	n/a
Chlorotrifluoromethane or Refrigerant gas R-13	2.2	UN1022	n/a
Dichlorodifluoromethane or Refrigerant gas R-12	2.2	UN1028	n/a
Dichlorodifluoromethane and Difluoroethane azeotropic mixture or Refrigerant gas R-500	2.2	UN2602	n/a
Dichlorofluoromethane or Refrigerant gas R-21	2.2	UN1029	n/a
Hexafluoroethane, compressed or Refrigerant gas R-116	2.2	UN2193	n/a
Hexafluoropropylene, or Refrigerant gas R-1216	2.2	UN1858	n/a
Tetrafluoromethane or Refrigerant gas R-14	2.2	UN1982	n/a